

David Z. Pan

professor

基本信息

姓名: David Z. Pan 所属院校: 德克萨斯州大学-奥斯汀
地区: United States 企业经历: 有

荣誉称号

★ ACM Fellow

★ SPIE Fellow

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

电子与计算机工程

微电子科学与工程

研究领域

EDA工具与电路验证

集成电路设计与制造

计算机体系结构

超大规模集成电路设计

数字集成电路设计

低功耗电路与能量管理

混合信号集成技术

模拟与射频IC设计

电路与系统设计

功率器件物理与结构

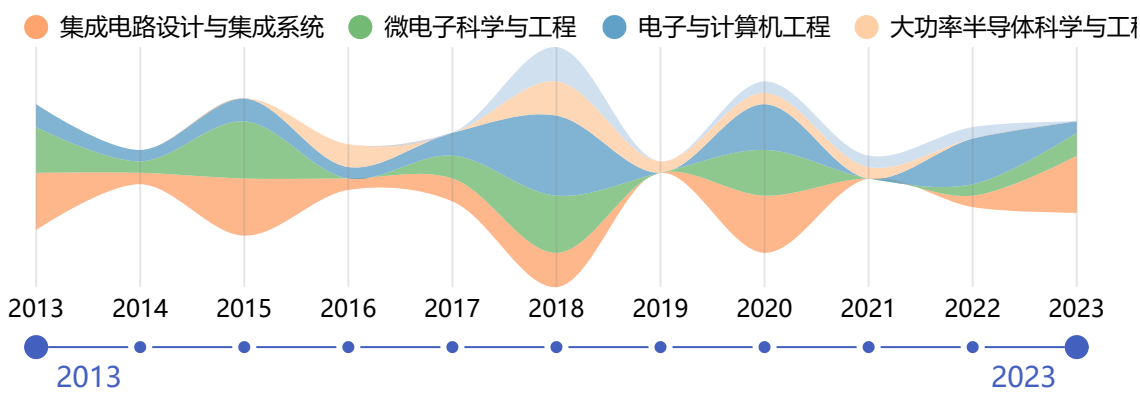
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个人简介

David Z. Pan is a Fellow of ACM, IEEE, and SPIE, and holds the Silicon Labs Endowed Chair as a Professor in the Department of Electrical and Computer Engineering at The University of Texas at Austin. He is also a GSC Member of the Computer Science Department. His research focuses on VLSI design automation, and he teaches courses including Introduction to VLSI Design and

VLSI I. His work has been widely recognized, with numerous accolades for both his students and collaborators.

近年发表的论文



A Bandwidth-Adaptive Pipelined SAR ADC With Three-Stage Cascoded Floating Inverter Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tang, Xiyuan; Yang, Xiangxing; Liu, Jiaxin; Wang, Zongnan; Shi, Wei; Pan, David Z.; Sun, Nan

被引用次数: 32023

Interactive Analog Layout Editing With Instant Placement and Routing Legalization

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gao, Xiaohan; Zhang, Haoyi; Liu, Mingjie; Shen, Linxiao; Pan, David Z.; Lin, Yibo; Wang, Runsheng; Huang, Ru

被引用次数: 02023

PISA: A Non-Volatile Processing-in-Sensor Accelerator for Imaging Systems

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Angizi, Shaahin; Tabrizchi, Sepehr; Pan, David Z.; Roohi, Arman

被引用次数: 12023

SqueezeLight: A Multi-Operand Ring-Based Optical Neural Network With Cross-Layer Scalability

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gu, Jiaqi; Feng, Chenghao; Zhu, Hanqing; Zhao, Zheng; Ying, Zhoufeng; Liu, Mingjie; Chen, Ray T.; Pan, David Z.

被引用次数: 62023

Hierarchical Analog and Mixed-Signal Circuit Placement Considering System Signal Flow

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhu, Keren; Chen, Hao; Liu, Mingjie; Pan, David Z.

被引用次数: 22023

Tutorial and Perspectives on MAGICAL: A Silicon-Proven Open-Source Analog IC Layout System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhu, Keren; Chen, Hao; Liu, Mingjie; Pan, David Z.

被引用次数: 6

2023

elfPlace: Electrostatics-Based Placement for Large-Scale Heterogeneous FPGAs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Meng, Yibai; Li, Wuxi; Lin, Yibo; Pan, David Z.

被引用次数: 11

2022

MLCAD: A Survey of Research in Machine Learning for CAD Keynote Paper

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Rapp, Martin; Amrouch, Hussam; Lin, Yibo; Yu, Bei; Pan, David Z.; Wolf, Marilyn; Henkel, Joerg

被引用次数: 26

2022

A Broadband Spectrum Channelizer With PWM-LO-Based Sub-Band Gain Control

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Ki Yong; Pan, David Z.; Gharpurey, Ranjit

被引用次数: 2

2022

An Efficient Analog Circuit Sizing Method Based on Machine Learning Assisted Global Optimization

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Budak, Ahmet Faruk; Gandara, Miguel; Shi, Wei; Pan, David Z.; Sun, Nan; Liu, Bo

被引用次数: 30

2022

A Compact Butterfly-Style Silicon Photonic-Electronic Neural Chip for Hardware-Efficient Deep Learning

ACS PHOTONICS

Feng, Chenghao; Gu, Jiaqi; Zhu, Hanqing; Ying, Zhoufeng; Zhao, Zheng; Pan, David Z.; Chen, Ray T.

被引用次数: 19

2022

Light in AI: Toward Efficient Neurocomputing With Optical Neural Networks-A Tutorial

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Gu, Jiaqi; Feng, Chenghao; Zhu, Hanqing; Chen, Ray T.; Pan, David Z.

被引用次数: 11

2022

An Efficient Automatic Structure Design Method of Silicon-on-Insulator Lateral Power Device Considering RESURF Constraint

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chen, Jing; Guo, Yufeng; Alawieh, Mohamed Baker; Zhang, Maolin; Zhang, Jun; Pan, David Z.

被引用次数: 4

2021

Toward High-Speed and Energy-Efficient Computing: A WDM-Based Scalable On-Chip Silicon Integrated Optical Comparator

LASER & PHOTONICS REVIEWS

Feng, Chenghao; Ying, Zhoufeng; Zhao, Zheng; Gu, Jiaqi; Pan, David Z.; Chen, Ray T.

被引用次数: 10

2021

Sequential Logic and Pipelining in Chip-Based Electronic-Photonic Digital Computing

IEEE PHOTONICS JOURNAL

Ying, Zhoufeng; Feng, Chenghao; Zhao, Zheng; Gu, Jiaqi; Soref, Richard; Pan, David Z.; Chen, Ray T.

被引用次数: 6

2020

A 13.5-ENOB, 107- μ W Noise-Shaping SAR ADC With PVT-Robust Closed-Loop Dynamic Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tang, Xiyuan; Yang, Xiangxing; Zhao, Wenda; Hsu, Chen-Kai; Liu, Jiaxin; Shen, Linxiao; Mukherjee, Abhishek; Shi, Wei; Li, Shaolan; Pan, David Z.; Sun, Nan

被引用次数: 50

2020

An Energy-Efficient Time-Domain Incremental Zoom Capacitance-to-Digital Converter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tang, Xiyuan; Li, Shaolan; Yang, Xiangxing; Shen, Linxiao; Zhao, Wenda; Williams, Randall P.; Liu, Jiaxin; Tan, Zhichao; Hall, Neal A.; Pan, David Z.; Sun, Nan

被引用次数: 23

2020

Electronic-photonic arithmetic logic unit for high-speed computing

NATURE COMMUNICATIONS

Ying, Zhoufeng; Feng, Chenghao; Zhao, Zheng; Dhar, Shounak; Dalir, Hamed; Gu, Jiaqi; Cheng, Yue; Soref, Richard; Pan, David Z.; Chen, Ray T.

被引用次数: 94

2020

Wavelength-division-multiplexing (WDM)-based integrated electronic-photonic switching network (EPSN) for high-speed data processing and transportation

NANOPHOTONICS

Feng, Chenghao; Ying, Zhoufeng; Zhao, Zheng; Gu, Jiaqi; Pan, David Z.; Chen, Ray T.

被引用次数: 17

2020

Automatic Selection of Structure Parameters of Silicon on Insulator Lateral Power Device Using Bayesian Optimization

IEEE ELECTRON DEVICE LETTERS

Chen, Jing; Alawieh, Mohamed Baker; Lin, Yibo; Zhang, Maolin; Zhang, Jun; Guo, Yufeng; Pan, David Z.

被引用次数: 13

2020

ABCDPlace: Accelerated Batch-Based Concurrent Detailed Placement on Multithreaded CPUs and GPUs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lin, Yibo; Li, Wuxi; Gu, Jiaqi; Ren, Haoxing; Khailany, Bruce; Pan, David Z.

被引用次数: 26

2020

An OTA-Less Second-Order VCO-Based CT $\Delta\Sigma$ Modulator Using an Inherent Passive Integrator and Capacitive Feedback

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Shaolan; Pan, David Z.; Sun, Nan

被引用次数: 17

2020

An Energy-Efficient Comparator With Dynamic Floating Inverter Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tang, Xiyuan; Shen, Linxiao; Kasap, Begum; Yang, Xiangxing; Shi, Wei; Mukherjee, Abhishek; Pan, David Z.; Sun, Nan

被引用次数: 105

2020

A 0.025-mm² 0.8-V 78.5-dB SNDR VCO-Based Sensor Readout Circuit in a Hybrid PLL- $\Delta\Sigma$ Structure

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Wenda; Li, Shaolan; Xu, Biying; Yang, Xiangxing; Tang, Xiyuan; Shen, Linxiao; Lu, Nanshu; Pan, David Z.; Sun, Nan

被引用次数: 42

2020

Semisupervised Hotspot Detection With Self-Paced Multitask Learning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chen, Ying; Lin, Yibo; Gai, Tianyang; Su, Yajuan; Wei, Yayi; Pan, David Z.

被引用次数: 26

2020

TimingCamouflage plus : Netlist Security Enhancement With Unconventional Timing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Grace Li; Li, Bing; Li, Meng; Yu, Bei; Pan, David Z.; Brunner, Michaela; Sigl, Georg; Schlichtmann, Ulf

被引用次数: 7

2020

Analysis of Microresonator-Based Logic Gate for High-Speed Optical Computing in Integrated Photonics

IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS

Feng, Chenghao; Ying, Zhoufeng; Zhao, Zheng; Mital, Rohan; Pan, David Z.; Chen, Ray T.

被引用次数: 5

2020

A New Physical Understanding of Lateral Step Doping Technique via Effective Concentration Profile Concept

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Jun; Guo, Yu-Feng; Pan, David Z.; Hu, Fang-Ren

被引用次数: 4

2019

A New Low Turn-Off Loss SOI Lateral Insulated Gate Bipolar Transistor With Buried Variation of Lateral Doping Layer

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Tian, Tao; Zhang, Sheng-Li; Guo, Yu-Feng; Zhang, Jun; Pan, David Z.; Yang, Ke-Meng

被引用次数: 5

2019

On the Approximation Resiliency of Logic Locking and IC Camouflaging Schemes

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Shamsi, Kaveh; Meade, Travis; Li, Meng; Pan, David Z.; Jin, Yier

被引用次数: 59

2019

UTPlaceF: A Routability-Driven FPGA Placer With Physical and Congestion Aware Packing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Li, Wuxi; Dhar, Shounak; Pan, David Z.

被引用次数: 44

2018

Thermal Stress and Reliability Analysis of TSV-Based 3-D ICs With a Novel Adaptive Strategy Finite Element Method

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhou, Hao; Zhu, Hengliang; Cui, Tao; Pan, David Z.; Zhou, Dian; Zeng, Xuan

被引用次数: 8

2018

TILA-S: Timing-Driven Incremental Layer Assignment Avoiding Slew Violations

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Liu, Derong; Yu, Bei; Chowdhury, Salim; Pan, David Z.

被引用次数: 17

2018

Clock Network Optimization With Multibit Flip-Flop Generation Considering Multicorner Multimode Timing Constraint

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lee, Taehee; Pan, David Z.; Yang, Joon-Sung

被引用次数: 6

2018

A 13-ENOB Second-Order Noise-Shaping SAR ADC Realizing Optimized NTF Zeros Using the Error-Feedback Structure

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Shaolan; Qiao, Bo; Gandara, Miguel; Pan, David Z.; Sun, Nan

被引用次数: 93

2018

MrDP: Multiple-Row Detailed Placement of Heterogeneous-Sized Cells for Advanced Nodes

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lin, Yibo; Yu, Bei; Xu, Xiaoqing; Gao, Jih-Rong; Viswanathan, Natarajan; Liu, Wen-Hao; Li, Zhuo; Alpert, Charles J.; Pan, David Z.

被引用次数: 25

2018

Report on the 2017 International Conference on Computer-Aided Design (ICCAD)

IEEE DESIGN & TEST

Parameswaran, Sri; Bahar, R. Iris; Pan, David Z.

被引用次数: 0

2018

Electro-Optic Ripple-Carry Adder in Integrated Silicon Photonics for Optical Computing

IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS

Ying, Zhouneng; Dhar, Shounak; Zhao, Zheng; Feng, Chenghao; Mital, Rohan; Chung, Chi-Jui; Pan, David Z.; Soref, Richard A.; Chen, Ray T.

被引用次数: 18

2018

Layout Synthesis for Topological Quantum Circuits With 1-D and 2-D Architectures

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lin, Yibo; Yu, Bei; Li, Meng; Pan, David Z.

被引用次数: 9

2018

Silicon microdisk-based full adders for optical computing

OPTICS LETTERS

Ying, Zhouneng; Wang, Zheng; Zhao, Zheng; Dhar, Shounak; Pan, David Z.; Soref, Richard; Chen, Ray T.

被引用次数: 53

2018

Interlayer Cooling Network Design for High-Performance 3D ICs Using Channel Patterning and Pruning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhu, Hengliang; Hu, Feng; Zhou, Hao; Pan, David Z.; Zhou, Dian; Zeng, Xuan

被引用次数: 7

2018

Cut Redistribution and Insertion for Advanced 1-D Layout Design via Network Flow Optimization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Ye; Lyu, Wenlong; Luk, Wai-Shing; Yang, Fan; Zhou, Hai; Zhou, Dian; Pan, David Z.; Zeng, Xuan

被引用次数: 0

2018

Effective Doping Concentration Theory: A New Physical Insight for the Double-RESURF Lateral Power Devices on SOI Substrate

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Jun; Guo, Yu-Feng; Pan, David Z.; Yang, Ke-Meng; Lian, Xiao-Juan; Yao, Jia-Fei

被引用次数: 20

2018

A Novel Variation of Lateral Doping Technique in SOI LDMOS With Circular Layout

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yang, Kemeng; Guo, Yufeng; Pan, David Z.; Zhang, Jun; Li, Man; Tong, Yi; He, Lin; Yao, Jiafei

被引用次数: 19

2018

A New Physical Insight for the 3-D-Layout-Induced Cylindrical Breakdown in Lateral Power Devices on SOI Substrate

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Jun; Guo, Yufeng; Pan, David Z.; Yang, Kemeng; Lian, Xiaojuan; Yao, Jiafei; Li, Man

被引用次数: 6

2018

Effective Concentration Profile: Mechanism of Gate Field-Plate Assistant Effect in SOI Lateral Power Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Jun; Guo, Yu-Feng; Pan, David Z.

被引用次数: 9

2018

Graph-Based Redundant Via Insertion and Guiding Template Assignment for DSA-MP

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Li, Xingquan; Yu, Bei; Ou, Jiaojiao; Chen, Jianli; Pan, David Z.; Zhu, Wenxing

被引用次数: 5

2018

Subresolution Assist Feature Generation With Supervised Data Learning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Xu, Xiaoqing; Lin, Yibo; Li, Meng; Matsunawa, Tetsuaki; Nojima, Shigeki; Kodama, Chikaaki; Kotani, Toshiya; Pan, David Z.

被引用次数: 16

2018

Automated logic synthesis for electro-optic logic-based integrated optical computing

OPTICS EXPRESS

Ying, Zhoufeng; Zhao, Zheng; Feng, Chenghao; Mital, Rohan; Dhar, Shounak; Pan, David Z.; Soref, Richard; Chen, Ray T.

被引用次数: 26

2018

Comparison of microrings and microdisks for high-speed optical modulation in silicon photonics

APPLIED PHYSICS LETTERS

Ying, Zhoufeng; Wang, Zheng; Zhao, Zheng; Dhar, Shounak; Pan, David Z.; Soref, Richard; Chen, Ray T.

被引用次数: 33

2018

Triple Patterning Aware Detailed Placement Toward Zero Cross-Row Middle-of-Line Conflict

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lin, Yibo; Yu, Bei; Xu, Biying; Pan, David Z.

被引用次数: 12

2017

Incremental Layer Assignment Driven by an External Signoff Timing Engine

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Livramento, Vinicius; Liu, Derong; Chowdhury, Salim; Yu, Bei; Xu, Xiaoqing; Pan, David Z.; Guntzel, Jose Luis; dos Santos, Luiz C. V.

被引用次数: 4

2017

Redundant Local-Loop Insertion for Unidirectional Routing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Xu, Xiaoqing; Lin, Yibo; Li, Meng; Ou, Jiaojiao; Cline, Brian; Pan, David Z.

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2017

High Performance Dummy Fill Insertion With Coupling and Uniformity Constraints

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lin, Yibo; Yu, Bei; Pan, David Z.

被引用次数: 6

2017

Polynomial Time Algorithm for Area and Power Efficient Adder Synthesis in High-Performance Designs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Roy, Subhendu; Choudhury, Mihir; Puri, Ruchir; Pan, David Z.

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2016

OSFA: A New Paradigm of Aging Aware Gate-Sizing for Power/Performance Optimizations Under Multiple Operating Conditions

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Roy, Subhendu; Liu, Derong; Singh, Jagmohan; Um, Junhyung; Pan, David Z.

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2016

Novel 3-D Analytical Method for Curvature Effect-Induced Electric Field Crowding in SOI Lateral Power Device

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Jun; Guo, Yu-Feng; Pan, David Z.; Yang, Ke-Meng

被引用次数: 8

2016

Layout Decomposition Co-Optimization for Hybrid E-Beam and Multiple Patterning Lithography

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Yang, Yunfeng; Luk, Wai-Shing; Pan, David Z.; Zhou, Hai; Yan, Changhao; Zhou, Dian; Zeng, Xuan

被引用次数: 11

2016

Self-aligned double-patterning layout decomposition for two-dimensional random metals for sub-10-nm node design

JOURNAL OF MICRO-NANOLITHOGRAPHY MEMS AND MOEMS

Ban, Yongchan; Pan, David Z.

被引用次数: 5

2015

Triple patterning lithography layout decomposition using end-cutting

JOURNAL OF MICRO-NANOLITHOGRAPHY MEMS AND MOEMS

Yu, Bei; Roy, Subhendu; Gao, Jhih-Rong; Pan, David Z.

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2015

Accurate lithography hotspot detection based on principal component analysis-support vector machine classifier with hierarchical data clustering

JOURNAL OF MICRO-NANOLITHOGRAPHY MEMS AND MOEMS

Yu, Bei; Gao, Jhih-Rong; Ding, Duo; Zeng, Xuan; Pan, David Z.

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2015

Directed self-assembly cut mask assignment for unidirectional design

JOURNAL OF MICRO-NANOLITHOGRAPHY MEMS AND MOEMS

Ou, Jiaojiao; Yu, Bei; Gao, Jhih-Rong; Pan, David Z.

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2015

Layout Decomposition for Triple Patterning Lithography

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Yu, Bei; Yuan, Kun; Ding, Duo; Pan, David Z.

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2015

Clock Tree Resynthesis for Multi-Corner Multi-Mode Timing Closure

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Roy, Subhendu; Mattheakis, Pavlos M.; Masse-Navette, Laurent; Pan, David Z.

被引用次数: 23

2015

Methodology for Standard Cell Compliance and Detailed Placement for Triple Patterning Lithography

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Yu, Bei; Xu, Xiaoqing; Gao, Jih-Rong; Lin, Yibo; Li, Zhuo; Alpert, Charles J.; Pan, David Z.

被引用次数: 31

2015

Self-Aligned Double Patterning Aware Pin Access and Standard Cell Layout Co-Optimization

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Xu, Xiaoqing; Cline, Brian; Yeric, Greg; Yu, Bei; Pan, David Z.

被引用次数: 46

2015

TSV Stress-Aware Full-Chip Mechanical Reliability Analysis and Optimization for 3D IC COMMUNICATIONS OF THE ACM

Jung, Moongon; Mitra, Joydeep; Pan, David Z.; Lim, Sung Kyu

被引用次数: 59

2014

Towards Optimal Performance-Area Trade-Off in Adders by Synthesis of Parallel Prefix Structures

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Roy, Subhendu; Choudhury, Mihir; Puri, Ruchir; Pan, David Z.

被引用次数: 28

2014

Electromigration Study for Multiscale Power/Ground Vias in TSV-Based 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Pak, Jiwoo; Lim, Sung Kyu; Pan, David Z.

被引用次数: 14

2014

Structure-Aware Placement Techniques for Designs With Datapaths

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ward, Samuel I.; Kim, Myung-Chul; Viswanathan, Natarajan; Li, Zhuo; Alpert, Charles J.; Swartzlander, Earl E., Jr.; Pan, David Z.

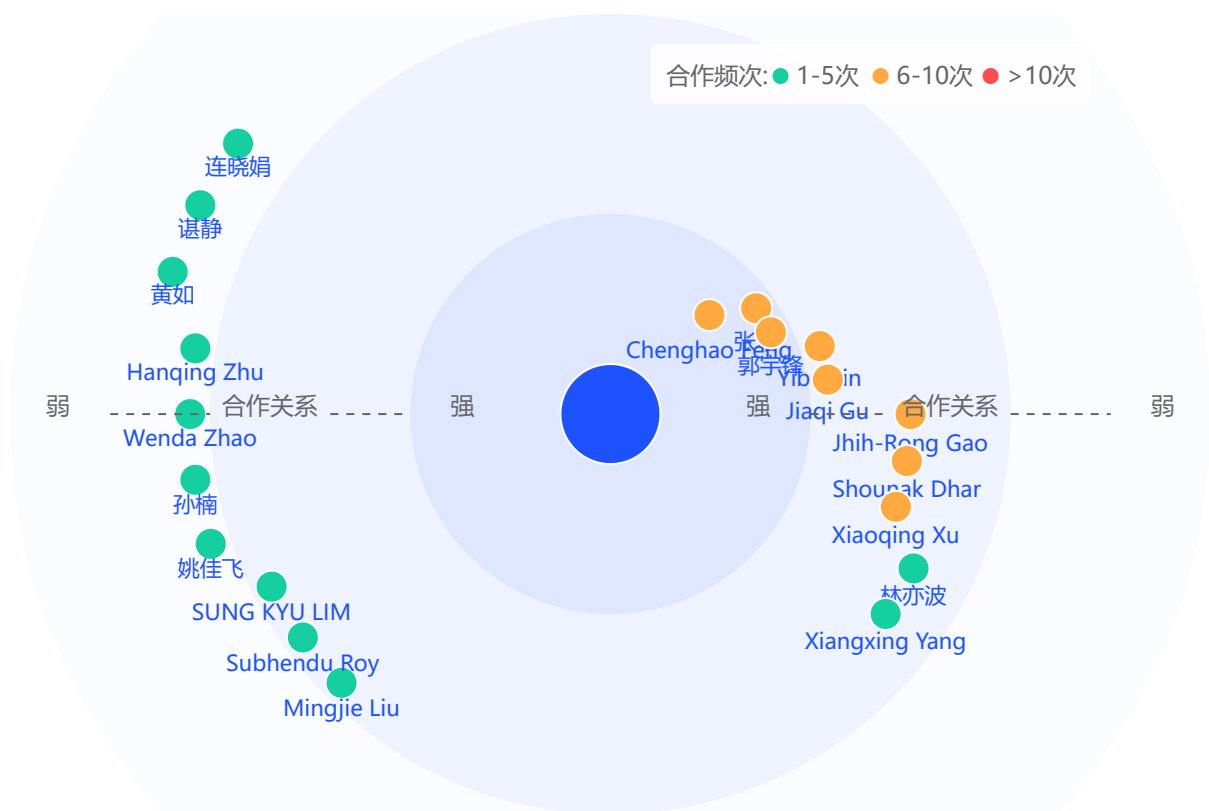
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2013

Chemical-Mechanical Polishing-Aware Application-Specific 3D NoC Design

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS Jang, Wooyoung; Pan, David Z. 被引用次数: 2	2013
Skew Management of NBTI Impacted Gated Clock Trees IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS Chakraborty, Ashutosh; Pan, David Z. 被引用次数: 8	2013
Impact of Mechanical Stress on the Full Chip Timing for Through-Silicon-Via-based 3-D ICs IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS Athikulwongse, Krit; Yang, Jae-Seok; Pan, David Z.; Lim, Sung Kyu 被引用次数: 15	2013
Design for Manufacturing With Emerging Nanolithography IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS Pan, David Z.; Yu, Bei; Gao, Jhih-Rong 被引用次数: 92	2013
Chip/Package Mechanical Stress Impact on 3-D IC Reliability and Mobility Variations IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS Jung, Moongon; Pan, David Z.; Lim, Sung Kyu 被引用次数: 15	2013
Investigations on Line-Edge Roughness (LER) and Line-Width Roughness (LWR) in Nanoscale CMOS Technology: Part II-Experimental Results and Impacts on Device Variability IEEE TRANSACTIONS ON ELECTRON DEVICES Wang, Runsheng; Jiang, Xiaobo; Yu, Tao; Fan, Jiewen; Chen, Jiang; Pan, David Z.; Huang, Ru 被引用次数: 54	2013

学术合作



Chenghao Feng

德克萨斯州大学-奥斯汀

合作次数：10

张琨

南京邮电大学

合作次数：9

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合作次数：8

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亚利桑那州立大学

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还有 50 个合作者